FAITHFUL FLATNESS, MAXIMAL IDEALS AND RESIDUE RINGS OF HENSELIZATION AND STRICT HENSELIZATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- The henselization is a colimit of etale algebras, and the strict henselization is a colimit of finite etale algebras. Algebra, Lemma \ref{algebra-lemma-henselization} and Algebra, Lemma \ref{algebra-lemma-strict-henselization}, cited at proof lines 10927-10929 and 10936-10937. Give the presentations from which flatness is read off.
- Etale maps are flat, flatness is stable under colimits, and a flat local homomorphism is faithfully flat. Algebra, Lemma \ref{algebra-lemma-etale}, Algebra, Lemma \ref{algebra-lemma-colimit-flat}, and Algebra, Lemma \ref{algebra-lemma-local-flat-ff}, cited at proof lines 10928-10934. Produce faithful flatness of the two ring maps.
- A flat local ring map which is an isomorphism on residue fields makes the source a basis of the target as a module over each Artinian quotient. Algebra, Lemma \ref{algebra-lemma-local-artinian-basis-when-flat}, cited at proof lines 10946-10948 and 10951-10954. Gives both the equality of the quotients and the freeness statement.
- Base change of a flat ring map is flat. Algebra, Lemma \ref{algebra-lemma-flat-base-change}, cited at proof lines 10945-10947. Makes the induced map on the quotients by powers of the maximal ideal flat.

Source reference. Original lemma 07QM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9837–9837. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 04GN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5839–5839. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00U2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1096–1096. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 05UT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7527–7527. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00HR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 653–653. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 04GP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5840–5840. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 051F: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6533–6533. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00HI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 644–644. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $(R, \mathfrak{m}, \kappa)$ be a local ring. Then we have the following*

1. *$R \to R^h \to R^{sh}$ are faithfully flat ring maps,*
2. *$\mathfrak{m}R^h = \mathfrak{m}^h$ and $\mathfrak{m}R^{sh} = \mathfrak{m}^h R^{sh} = \mathfrak{m}^{sh}$,*
3. *$R/\mathfrak{m}^n = R^h/\mathfrak{m}^n R^h$ for all n,*
4. *there exist elements $x_i \in R^{sh}$ such that $R^{sh}/\mathfrak{m}^n R^{sh}$ is a free $R/\mathfrak{m}^n$-module on $x_i \bmod \mathfrak{m}^n R^{sh}$.*

Proof. By construction R^h is a colimit of étale R-algebras, see Algebra, Lemma 04GN. Since étale ring maps are flat (Algebra, Lemma 00U2) we see that R^h is flat over R by Algebra, Lemma 05UT. As a flat local ring homomorphism is faithfully flat (Algebra, Lemma 00HR) we see that $R \to R^h$ is faithfully flat. The ring map $R^h \to R^{sh}$ is a colimit of finite étale ring maps, see proof of Algebra, Lemma 04GP. Hence the same arguments as above show that $R^h \to R^{sh}$ is faithfully flat.

Part (2) follows from Algebra, Lemmas 04GN and 04GP. Part (3) follows from Algebra, Lemma 051F because $R/\mathfrak{m} \to R^h/\mathfrak{m}R^h$ is an isomorphism and $R/\mathfrak{m}^n \to R^h/\mathfrak{m}^n R^h$ is flat as a base change of the flat ring map $R \to R^h$ (Algebra, Lemma 00HI). Let κ^{sep} be the residue field of R^{sh} (it is a separable algebraic closure of κ). Choose $x_i \in R^{sh}$ mapping to a basis of κ^{sep} as a κ-vector space. Then (4) follows from Algebra, Lemma 051F in exactly the same way as above. □